

Assignment 1

Due Wednesday 01/21/26

Reading:

- Required: Lecture Notes 2
- Required: Lecture Slides 2
- Optional: ITILA 26

Problems:

1. (ECCMMA 16.4) Consider the functions

$$f(x_1, x_2, \dots, x_9) = f_1(x_1, x_3)f_2(x_1, x_4, x_5)f_3(x_1, x_6, x_7)f_4(x_4, x_8, x_9)$$

and

$$g_1(x_1) = \sum_{x_1^9 \setminus x_1} f(x_1, x_2, \dots, x_9).$$

Assume that each x_i comes from an alphabet $\mathcal{X}$ with $|\mathcal{X}| = q$.

- Determine the number of computations required to compute $g_1(x_1)$ for all $x_1 \in \mathcal{X}$ if the factorization is not used.
- Draw the factor graph associated with the given factorization of f .
- What is the complexity of computing $g_1(x_1)$ for all $x_1 \in \mathcal{X}$ with message passing if only the relevant messages are computed? Determine the messages passed on each edge of the tree for this computation.
- If only the relevant messages are computed, how many additional computations are required to compute

$$g_7(x_7) = \sum_{x_1^9 \setminus x_7} f(x_1, x_2, \dots, x_9)?$$

2. (ECCMMA 16.7) Let $P_{X_1, X_2, X_3, X_4}(x_1, x_2, x_3, x_4)$ be a joint probability mass function for the random variables X_1, X_2, X_3, X_4 .

- Draw the (un)factor graph representing $P_{X_1, \dots, X_4}(x_1, x_2, x_3, x_4)$.
- Draw the factor graph representing standard decomposition

$$P_{X_1, \dots, X_4}(x_1, x_2, x_3, x_4) = P_{X_1}(x_1)P_{X_2|X_1}(x_2|x_1)P_{X_3|X_1, X_2}(x_3|x_1, x_2)P_{X_4|X_1, X_2, X_3}(x_4|x_1, x_2, x_3).$$

- Assume $X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow X_4$ form a Markov chain and draw the factor graph associated with the factorization

$$P_{X_1, \dots, X_4}(x_1, x_2, x_3, x_4) = P_{X_1}(x_1)P_{X_2|X_1}(x_2|x_1)P_{X_3|X_2}(x_3|x_2)P_{X_4|X_3}(x_4|x_3).$$

- (d) Suppose that the X_i are observed indirectly through $Y_i \sim P_{Y_i|X_i}(y|x) = p(y_i|x_i)$. Draw the factor graph for the hidden Markov model (HMM) implied by the factorization

$$P_{X_1, \dots, X_4, Y_1, \dots, Y_4}(x_1, \dots, x_4, y_1, \dots, y_4) = \prod_{i=1}^4 P_{X_i|X_{i-1}}(x_i|x_{i-1})p(y_i|x_i),$$

where it is assumed that $P_{X_1|X_0}(x_1|x_0) = P_{X_1}(x_1)$ (e.g., X_1 is independent of X_0).

3. (FGML 1.3) Once upon a time, your friendly neighborhood TA was intent on world domination, but now she just wants to graduate. Her first step, obviously, was to build a graphical model, as shown below. The variables being: Graduate (G), which depends on papers (P) and proposal (R). As we know, papers and theorems (T) are both generated by drinking lots of coffee (C). And all grad students are dependent on free coffee provided by the university (U), which also allows depends on their favorite coffee shop being open (D). All the variables are binary valued $\{\text{T}, \text{F}\}$. The CPT parameters are:

$$P(U = \text{T}) = 0.1 \tag{1}$$

$$P(D = \text{T} \mid U = \text{T}) = 0.6, \quad P(D = \text{T} \mid U = \text{F}) = 0.5 \tag{2}$$

$$P(C = \text{T} \mid U = \text{T}, D = \text{T}) = 0.7, \quad P(C = \text{T} \mid U = \text{T}, D = \text{F}) = 0.5$$

$$P(C = \text{T} \mid U = \text{F}, D = \text{T}) = 0.6, \quad P(C = \text{T} \mid U = \text{F}, D = \text{F}) = 0.05 \tag{3}$$

$$P(P = \text{T} \mid C = \text{T}) = 0.8, \quad P(P = \text{T} \mid C = \text{F}) = 0.6 \tag{4}$$

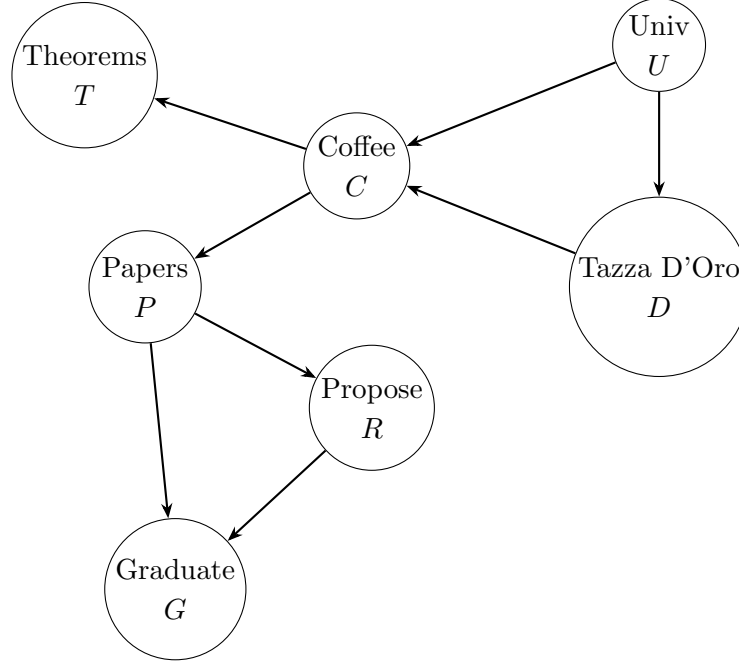
$$P(T = \text{T} \mid C = \text{T}) = 0.7, \quad P(T = \text{T} \mid C = \text{F}) = 0.6 \tag{5}$$

$$P(R = \text{T} \mid P = \text{T}) = 0.7, \quad P(R = \text{T} \mid P = \text{F}) = 0.1 \tag{6}$$

$$P(G = \text{T} \mid P = \text{T}, R = \text{T}) = 0.9, \quad P(G = \text{T} \mid P = \text{T}, R = \text{F}) = 0.3$$

$$P(G = \text{T} \mid P = \text{F}, R = \text{T}) = 0.5, \quad P(G = \text{T} \mid P = \text{F}, R = \text{F}) = 0.1 \tag{7}$$

Help your TA make some urgent inferences about her graduation plans.



- (a) How likely is the TA to graduate if her coffee supply runs out? $P(G = \text{T} \mid C = \text{F}) = ?$
- (b) How likely is the TA to graduate if Tazzo is open all the time? $P(G = \text{T} \mid D = \text{T}) = ?$
- (c) Should we even be worried about her coffee supply running out? $P(C = \text{T}) = ?$
- (d) If she is writing theorems, does this mean she will also publish? $P(P = \text{T} \mid T = \text{T}) = ?$

Additionally, report the ordering used and the factors produced after eliminating each variable for the first query $[P(G = \text{T} \mid C = \text{F})]$.

4. (MCT 2.4) (HANSEL AND GRETEL TAKE A FIELD TRIP IN THE DARK FOREST). Hansel and Gretel, together with their classmates, take a field trip. The forest in which they are walking is so dark that each child can only see his/her immediate neighbors. Assume that communication is limited to these nearest neighbors as well and that the whole group of schoolchildren forms a tree (in the graph sense) with respect to this neighborhood structure. Construct a message-passing algorithm that allows the children to count all their classmates in a distributed fashion and determine if any children were eaten by the wolf. What is the initialization and what are the message-passing rules? Can you modify the algorithm to only count a prescribed subset, e.g., the set of girls?
5. (FGML 1.5) Consider an undirected graph $G = (\mathcal{V}, \mathcal{E})$ with $\mathcal{V} = \{1, \dots, n\}$ and edges $\mathcal{E}$. An *independent set* is a subset $S \subseteq \mathcal{V}$ such that no edge has both endpoints in S . In this problem, you will model independent sets using a *pairwise factor graph* and use message passing ideas on a tree.

- (a) *Binary encoding.* Introduce binary variables $X_i \in \{0, 1\}$ for each $i \in \mathcal{V}$ where $X_i = 1$ means “vertex i is chosen.” Write a pairwise factor-graph representation of the constraint “ X encodes an independent set” using *only* pairwise factors over edges.

More concretely, define factors $\psi_{ij}(x_i, x_j)$ for each $(i, j) \in \mathcal{E}$ so that

$$f(x) = \prod_{(i,j) \in \mathcal{E}} \psi_{ij}(x_i, x_j)$$

is equal to 1 if $x = (x_1, \dots, x_n)$ is an independent-set indicator and 0 otherwise. Give an explicit formula for ψ_{ij} .

- (b) *Counting independent sets as a partition function.* Using the factors from part (a), define a distribution

$$p(x) = \frac{1}{Z} \prod_{(i,j) \in \mathcal{E}} \psi_{ij}(x_i, x_j),$$

and explain why the normalizing constant

$$Z = \sum_{x \in \{0,1\}^n} \prod_{(i,j) \in \mathcal{E}} \psi_{ij}(x_i, x_j)$$

counts the number of independent sets of G .

- (c) *The weighted (hard-core) variant.* Fix a parameter $\lambda > 0$ and define unary factors

$$\phi_i(x_i) = \lambda^{x_i} \quad (i \in \mathcal{V}).$$

Consider the weighted model

$$p_\lambda(x) = \frac{1}{Z(\lambda)} \left(\prod_{i \in \mathcal{V}} \phi_i(x_i) \right) \left(\prod_{(i,j) \in \mathcal{E}} \psi_{ij}(x_i, x_j) \right).$$

- i. Interpret $Z(\lambda)$ as a generating function for independent sets by size.
 - ii. Give an expression for $\mathbb{E}_{p_\lambda}[\sum_{i \in \mathcal{V}} X_i]$ in terms of $Z(\lambda)$ and λ (hint: differentiate $\log Z(\lambda)$).
- (d) *Exact computation on a small tree.* Consider the tree T shown in Figure 1. Use the pairwise-factor formulation from parts 1–3 with the weighted hard-core model parameter $\lambda > 0$.

- (a) *Set up sum-product messages.* Root the tree at vertex 1 and direct messages toward the root. For an edge $i-j$, define the message from i to j as

$$m_{i \rightarrow j}(x_j) := \sum_{x_i \in \{0,1\}} \phi_i(x_i) \psi_{ij}(x_i, x_j) \prod_{k \in N(i) \setminus \{j\}} m_{k \rightarrow i}(x_i),$$

where $N(i)$ are the neighbors of i in T . Write out $m_{4 \rightarrow 2}$ and $m_{5 \rightarrow 2}$ explicitly (as two numbers each, one for $x_2 = 0$ and one for $x_2 = 1$).

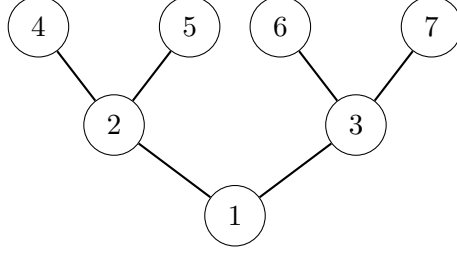


Figure 1: A small tree T with 7 vertices.

- (b) *Compute $Z(\lambda)$ exactly.* Compute all messages up to the root and then compute

$$Z(\lambda) = \sum_{x_1 \in \{0,1\}} \phi_1(x_1) \prod_{k \in N(1)} m_{k \rightarrow 1}(x_1).$$

Give your final answer as a polynomial in λ (simplified).

- (c) *Unweighted counting as a check.* Evaluate your expression at $\lambda = 1$ to obtain the number of independent sets of T .
- (d) *A marginal probability.* Compute the marginal $p_\lambda(X_1 = 1)$. Your final answer should be a rational function in λ .
- (e) *Sampling.* Describe how this could be used to efficiently sample from the set of independent sets.

Practice Problems (do not hand in):

- (MCT 2.2) (MESSAGE-PASSING ALGORITHM FOR BEC). Starting from the message-passing rules for marginalization on a factor graph, derive a simplified decoding algorithm for the binary erasure channel. What is the message alphabet and what are the computation rules? Simplify the rules as far as possible.
- (WEIGHT ENUMERATORS AND FACTOR GRAPHS). Consider a factor graph for binary variables that defines the decomposition

$$f(x_1, \dots, x_n) = \prod_{a=1}^M f_a(x_{\partial a}).$$

The weight enumerator function for this factor graph is defined to be

$$A(z) \triangleq \sum_{x_1^n \in \{0,1\}^n} f(x_1, \dots, x_n) z^{w_H(x_1^n)}.$$

- (a) How can we define local factors $f_i(x_i; z)$ for each $i \in [n]$ so that $A(z)$ is given by the partition function

$$Z \triangleq \sum_{x_1^n \in \{0,1\}^n} f(x_1, \dots, x_n) \prod_{i=1}^n f_i(x_i; z)?$$

- (b) For the Tanner graph on the right in the figure below, compute the weight enumerator efficiently using the distributive law.

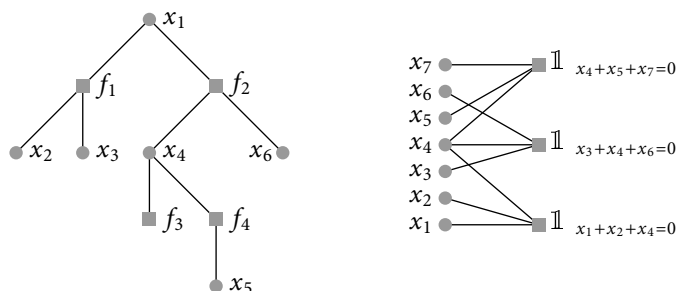
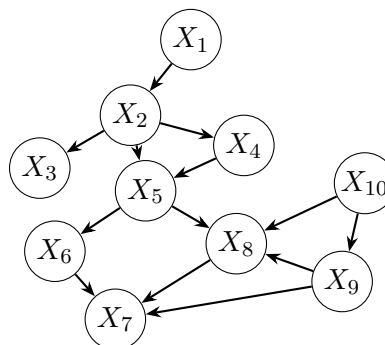


Figure 2.4: Left: Factor graph of f given in Example 2.2. Right: Factor graph for the code membership function defined in Example 2.5.

3. (FGML 1.2) For each of the following assertions of (conditional) independence, state if they are True or False with justification for the Bayesian network shown.

1. $X_1 \perp\!\!\!\perp X_9$
2. $X_{10} \perp\!\!\!\perp X_7 \mid X_9$
3. $X_2 \perp\!\!\!\perp X_4 \mid X_5$
4. $X_2 \perp\!\!\!\perp X_8 \mid X_5$
5. $X_5 \perp\!\!\!\perp X_{10} \mid X_7$
6. $X_5 \perp\!\!\!\perp X_{10} \mid X_8$
7. $X_1 \perp\!\!\!\perp X_9 \mid X_7$



4. (FGML 1.4) Two graduate students in Pittsburgh have gotten into an argument over the weather. One thinks summer is over and Autumn has already come, while the other thinks it is still summer.

In Pittsburgh, there are four seasons – Spring (S), Summer (M), Autumn (A), and Winter (W). Given the season the previous day, the season on a day is conditionally independent of the season on all previous days. The weather is either Hot (H), Rainy (R) or Freezing (F). Given the season on any given day, the weather that day is independent of all other variables.

More formally, we let C_i denote the season on the i -th day (taking values S, M, A, W) and O_i denote the observed weather pattern (one of H, R, F). Thus, we have $\forall j < i - 1$, $C_i \perp\!\!\!\perp C_j \mid C_{i-1}$ and $\forall X, O_i \perp\!\!\!\perp X \mid C_i$ where X is any random variable other than C_i, O_i .

1. (3 pts) Draw a factor graph model over $C_1, \dots, C_N, O_1, \dots, O_N$ that satisfies the conditional independencies listed above.

2. (7 pts) Implement sum-product and max-product algorithms for this graphical model.
3. (10 pts) We has made 20 observation of the weather over the last few months (i.e., $O_1, \dots, O_N$):

$$\{R, F, F, H, F, H, H, H, H, H, H, H, R, H, H, H, R, H, H\}.$$

Some of the values for the conditional probability table (CPT) are as follows.

$P(C_1)$:

	S	M	A	W
$P(C_1 = \cdot)$	0.15	0.6	0.2	0.05

$P(C_{t+1} = j \mid C_t = i)$ for all $t \geq 1$ (**i:row, j:column**):

	S	M	A	W
S	0.8	0.17	0.02	0.01
M	0.1	0.7	0.19	0.01
A	0.02	0.05	0.7	0.23
W	0.2	0.01	0.04	0.7

$P(O_t = j \mid C_t = i)$:

	H	R	F
S	0.4	0.3	0.3
M	0.5	0.45	0.05
A	0.3	0.4	0.3
W	0.0001	0.2499	0.75

For inference, apply both sum-product and max-product algorithms to the following problems. Submit all of your codes (zipped as `hw1_bp.zip`) and report the results.

- (a) Compute the probability of (S, M, A, W) for each of all 20 observations (e.g., $\forall t, P(C_t = M \mid O_1, \dots, O_N)$). Save the result of (4×20) probability matrix, and draw it into a figure (x-axis: 20 time steps, y-axis: probability).
- (b) Determine the most likely sequence of $C_1, \dots, C_N$ that generated this observed sequence.